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Initial evidence of research quality of registered reports compared with the standard publishing model

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In registered reports (RRs), initial peer review and in-principle acceptance occur before knowing the research outcomes. This combats publication bias and distinguishes planned from unplanned research. How RRs could improve the credibility of research findings is straightforward, but there is little empirical evidence. Also, there could be unintended costs such as reducing novelty. Here, 353 researchers peer reviewed a pair of papers from 29 published RRs from psychology and neuroscience and 57 non-RR comparison papers. RRs numerically outperformed comparison papers on all 19 criteria (mean difference 0.46, scale range −4 to +4) with effects ranging from RRs being statistically indistinguishable from comparison papers in novelty (0.13, 95% credible interval [−0.24, 0.49]) and creativity (0.22, [−0.14, 0.58]) to sizeable improvements in rigour of methodology (0.99, [0.62, 1.35]) and analysis (0.97, [0.60, 1.34]) and overall paper quality (0.66, [0.30, 1.02]). RRs could improve research quality while reducing publication bias and ultimately improve the credibility of the published literature.

Publication of journal articles is the primary mechanism for communicating the results of empirical research and a key incentive for researchers to advance in their careers. The standard model of peer review and publication is vulnerable to biases that can reduce the credibility of the published literature[1–5]. Some kinds of results are more likely to be published than others. Positive results—finding that a treatment is effective or that there is a relationship between two variables—are more likely to be published than negative results[6–12]. Novel or innovative results are more likely to be published than replications or incremental extensions of existing findings[13–15]. And, clean papers in which the evidence consistently supports the proposed conclusion are more likely to be published than papers with equivocal evidence, exceptions or other uncertainties[16,17]. Attention paid to positive, novel and clean findings could come at the expense of evaluating the quality and rigour of the research for making publication decisions. This may contribute to publication bias, selective reporting and questionable research practices that reduce the credibility of the published literature[18–25].

Registered reports (RRs) are designed to address these biases and improve the quality of research[1–3]. With RRs, the first stage of peer review occurs prior to knowing the research outcomes of the key study or studies, in most cases before data collection has occurred. Reviewers evaluate the importance of the research question and the quality of the methodology used to evaluate that question. The findings cannot distract reviewers from evaluating research quality and rigour because there are no findings yet. Expert review in advance should facilitate improving the methodology rather than just pointing out limitations of findings already obtained. If accepted, the authors receive an in-principle commitment from the journal to publish the results as long as the research is performed as proposed with evidence of competent execution and interpretation. Decisions to publish are therefore based on the perceived importance of the research question and the quality of the methodology, not whether the findings are positive, novel and clean.

More than 250 journals have adopted RRs since 2013 on the theorized promise of improving rigour and credibility. Initial evidence suggests that RRs are (1) effective at mitigating publication bias with a sharp increase in publishing negative results compared with the standard model[26,27], and (2) cited as often or even more than other articles in the same journals[28]. However, there is no evidence about whether scholars perceive RRs to have higher, lower or similar research quality compared with papers published in the standard model. The RR format could also have costs such as authors pursuing less interesting questions or conducting less novel or creative research[29,30].

We conducted an observational investigation of perceptions of the quality and importance of RRs compared with the standard model across a variety of outcome criteria. We recruited 353 researchers to each peer review a pair of papers, one from 29 RRs from psychology and neuroscience and one from 57 matched non-RR comparison papers. Comparison papers addressed similar topics, about half were by the same first or corresponding authors and about half were published in the same journal. RRs are a popular format for replication studies[3,31], but replications are rare in the standard model, so we excluded replication RRs. Researchers were assigned to papers according to their self-reported expertise based on the papers' keywords. Researchers self-reported that they were qualified to review the papers on average ($N = 353$; RR mean 3.74, s.d. 1.02; comparison paper mean 3.59, s.d. 1.07; range 1 [not at all qualified] to 5 [substantially qualified]). Reviewers evaluated 19 outcome criteria including quality, rigour, novelty, creativity and importance of the methodology and outcomes of the papers.

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In some RRs, authors submitted preliminary studies as initial evidence supporting the approach of the proposed last study that was peer reviewed before the findings were known. If the RR or non-RR had multiple studies (27 of 86), then the evaluation of methods and results considered the last study only. We also coded the papers on objective qualities of the research such as sample size, preregistration and data sharing. The findings provide an initial empirical test of the quality of research published using RRs compared with the standard model.

Results

Preregistered primary outcomes. Reviewers read and evaluated each paper in three steps. They first read and evaluated the introduction, initial studies (if any) and methodology of the last study before knowing the last study's results. Next, they read and evaluated the last study's results and discussion. Finally, they read the paper's abstract and provided final assessments of the paper. We preregistered analysis plans for comparing the difference between ratings of RRs and control articles on each criterion using a within-subjects comparison to maximize power and a between-subjects comparison for the article that reviewers were randomly assigned to read first. As estimated effects were similar across analysis strategies, we emphasize reporting the more precise within-subjects estimates. Our preregistered within-subjects model included a separate model for each outcome, but upon seeing the correlational structure of the outcomes (Extended Data Fig. 1), we decided to use a multi-level model including all outcomes with partial pooling to mitigate concerns regarding multiple comparisons.

The results of that model are presented in Fig. 1 and are consistent with the results of the preregistered within-subjects model. In the figure and paragraph below, we report point estimates and 95% credible intervals on the raw response scales. In text, we also report a standardized effect size metric and 95% credible intervals. Credible intervals do not have the same interpretation as the confidence intervals that are commonly reported in null-hypothesis significance testing. Given the evidence provided by the observed data and weakly informative priors, there is a 95% posterior probability that the true estimate lies within the credible interval. This is distinct from common uses of confidence intervals in null-hypothesis significance testing to declare a result 'statistically significant' based on whether the interval includes 0.

Before learning the findings of the last study, compared with control articles, reviewers evaluated the RRs' last study as having a more rigorous methodology (estimate 0.99, 95% credible interval [0.62, 1.35]; standardized effect size (ES) 0.43, 95% credible interval [0.27, 0.58]), higher quality methodology (estimate 0.77, [0.40, 1.12]; standardized ES 0.32, [0.17, 0.47]), higher estimates of what will be learned (estimate 0.22, [0.05, 0.38]; standardized ES 0.17, [0.01, 0.32]), better alignment between the research question and methodology (estimate 0.37, [0.01, 0.73]; standardized ES 0.19, [0.01, 0.37]) and higher quality research question (estimate 0.39, [0.03, 0.76]; standardized ES 0.20, [0.02, 0.38]). RRs were statistically indistinguishable from control articles in the importance of the research regardless of what results will be observed (estimate 0.27, [−0.09, 0.63]; standardized ES 0.12, [−0.04, 0.28]), creativity of the methodology (estimate 0.22, [−0.14, 0.58]; standardized ES 0.11, [−0.07, 0.28]) and novelty of the research questions (estimate 0.13, [−0.24, 0.49]; standardized ES 0.06, [−0.12, 0.25]).

After reading the results and discussion of the last study, compared with control articles, reviewers evaluated the RRs as having a more rigorous analysis strategy (estimate 0.96, [0.60, 1.34]; standardized ES 0.46, [0.28, 0.63]), more justified conclusions based on the results (estimate 0.65, [0.29, 1.01]; standardized ES 0.31, [0.14, 0.48]), higher quality of the results (estimate 0.61, [0.25, 0.97]; standardized ES 0.29, [0.12, 0.47]), higher quality of the discussion (estimate 0.40, [0.04, 0.77]; standardized ES 0.19, [0.02, 0.37]) and higher estimates of what was learned (estimate 0.40, [0.03, 0.77]; standardized ES 0.17, [0.01, 0.32]). RRs were statistically indistinguishable from control articles in whether the results were innovative (estimate 0.29, [−0.08, 0.65]; standardized ES 0.13, [−0.04, 0.30]) and the importance of the findings (estimate 0.28, [-0.08, 0.64]; standardized ES 0.12, [−0.04, 0.28]).

After finishing the paper, compared with control articles, reviewers evaluated the RRs as having higher overall quality (estimate 0.66, [0.30, 1.02]; standardized ES 0.32, [0.15, 0.50]) and producing more important discoveries (estimate 0.38, [0.02, 0.75]; standardized ES 0.17, [0.01, 0.34]). RRs were statistically indistinguishable from control articles in alignment between abstract and findings (estimate 0.33, [−0.03, 0.69]; standardized ES 0.21, [−0.02, −0.44]) and the likelihood to inspire new research (estimate 0.18, [−0.19, 0.54]; standardized ES 0.08, [−0.08, 0.24]).

In the aggregate, RRs numerically outperformed comparison articles on all 19 criteria with an average magnitude of 0.46 and a range of 0.13–0.96. Twelve of the 19 criteria showed performance advantages with 95% credible intervals not including zero.

We repeated the analysis as a between-subjects comparison using only the randomly assigned first reviewed article. This substantially reduces precision compared with the within-subjects design, but has the advantage of a possibly more 'pure' assessment of articles before reviewers might have noticed and compared differences between articles in the within-subjects design, potentially exaggerating observed effects. As with the within-subjects analysis, our originally preregistered model included a separate model for each outcome, but these results were similar to a model that included all outcomes (Supplementary Information). The observed effects from the between-subject model with all outcomes showed a similar pattern with an average effect size of 0.43 and a range of 0.04–0.93 (Supplementary Table 1).

Characteristics of papers and reviewers. When preparing the papers for review, we removed explicit mention of RRs from the text, but we did not otherwise attempt to blind reviewers from information about the methodology because their task was to evaluate the methodology. After completing the reviews, we informed reviewers about the study's focus on RRs and preregistration and asked about their experience and opinions. Less than half (47%; 154/326) had ever preregistered a study, and 14% (41/299) had submitted a RR. Even so, reviewers reported being relatively familiar with preregistration (mean 3.90; s.d. 1.22; range 1 [not at all] to 5 [substantially]) and RRs (mean 3.24, s.d. 1.33). Reviewers also reported believing that RRs can improve research rigour (mean 2.01, s.d. 1.32; range −4 [substantially worsen] to 4 [substantially improve]) and quality (mean 1.71, s.d. 1.53). Finally, a majority of reviewers correctly identified the RR as an RR (54%; 190 correct, 38 incorrect, 125 unsure) and the comparison paper as not an RR (54%; 192 correct, 30 incorrect, 131 unsure).

Qualification of outcomes. We conducted exploratory analyses of how characteristics and beliefs of the reviewers might qualify the performance outcomes for RRs compared with other papers. Our sample was positive towards RRs and familiar with the format with some variation, so we explored how that may have affected our results. Beliefs about quality and rigour of RRs were highly correlated ($r=0.82$; 95% confidence interval [0.78, 0.85]), as were familiarity with RR and preregistration ($r=0.74$; 95% confidence interval [0.69, 0.79]), so we averaged these questions to form single familiarity and improvement belief items. To allow for a non-linear relationship, we kept the resulting averages as categorical variables, rounding half values to create a five- and six-level factor, respectively. These were then included as random effects in two within-subjects models to allow for partial pooling across factor levels and dependent variables to mitigate concerns regarding multiple comparisons[32]. Those

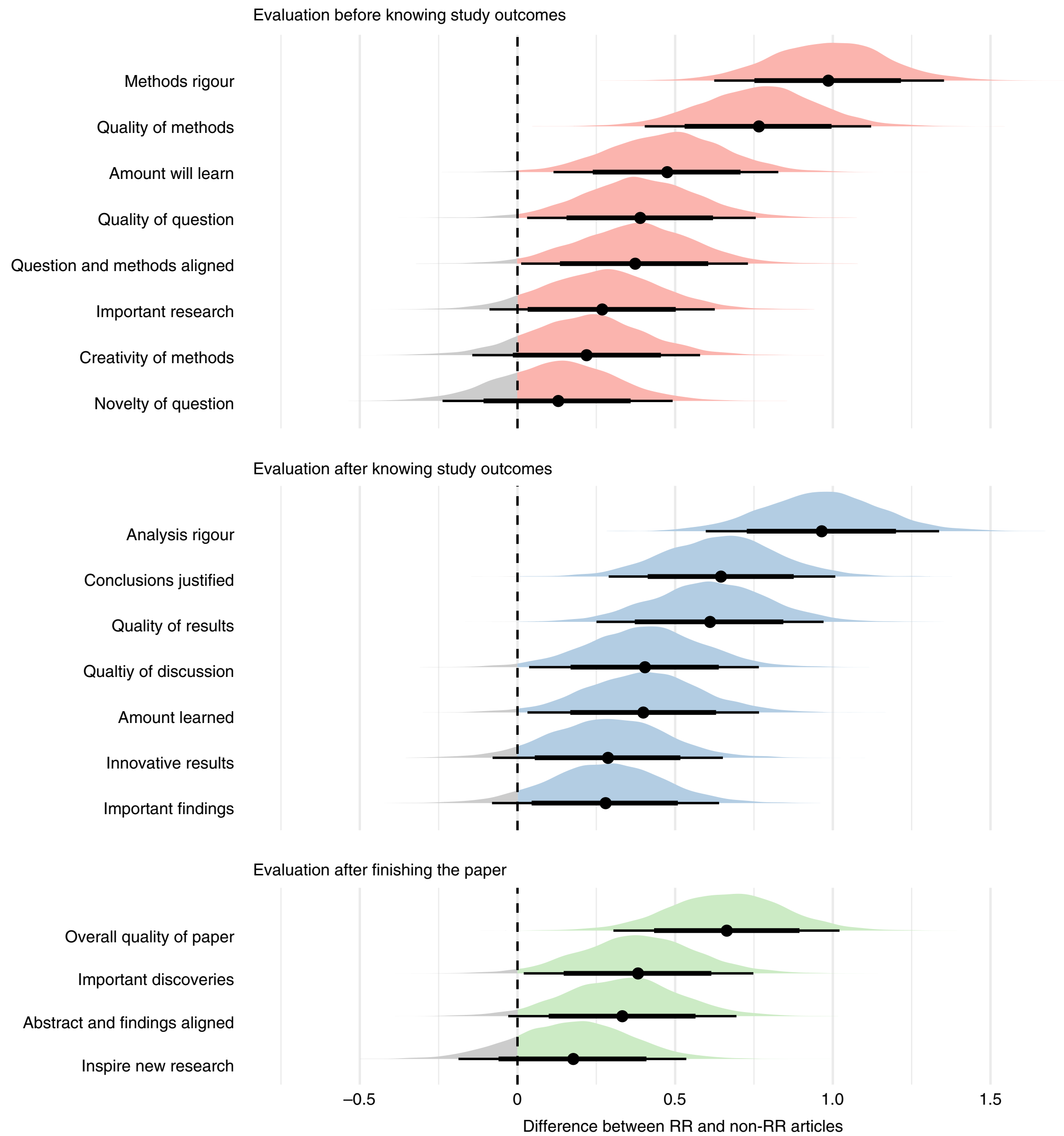

Fig. 1 | Posterior probability distributions for parameter estimates from within-subjects analysis using partial pooling across the 19 outcomes comparing the difference of RRs and comparison articles. Black horizontal lines indicate 80% (thick) and 95% (thin) credible intervals, and dots show the mean of the posteriors. Positive values indicate a performance advantage for RRs; negative values indicate a performance advantage for comparison articles. Outcomes are sorted from strongest to weakest performance advantage for RRs in the three evaluation groups: before knowing study outcomes, after knowing study outcomes and after finishing the paper.

reporting the highest (mean 0.54, 95% credible interval [0.12, 0.99]) and next highest level of familiarity (0.45, [0.06, 0.83]) perceived a larger performance advantage for RRs than those reporting the lowest level of familiarity (0.32, [−0.24, 0.77]) but similarly with the second lowest level of familiarity (0.49, [0.06, 0.95]) and the middle level of familiarity (0.42, [0.02, 0.81]; see Extended Data Fig. 2 and Supplementary Table 2 for more details). And, unsurprisingly, those reporting very strong (0.82, [0.26, 1.40]), strong (0.52, [0.11, 0.94]) and moderately strong beliefs that RRs improve rigour (0.46, [0.03, 0.89]) showed a larger performance advantage for RRs those who were slightly favourable toward RRs (0.26, [−0.20, 0.71]), neutral toward RRs (0.45, [−0.07, 0.97]) or believed RRs actually decreased quality and rigour (0.24, [−0.45, 0.82]; see Extended Data Fig. 3 and Supplementary Table 3 for more details). It is not surprising that believing that the RR format improves rigour and quality would translate to higher ratings of rigour and quality when rating individual RRs, just as it would not be surprising that believing that random assignment, random sampling and blinding improve quality would translate to higher ratings when those features were present versus absent. Notably, however, every subgroup estimate was in the direction of a performance advantage for RRs, even among those reviewers who believed that RRs were neutral or negative for enhancing rigour and quality in general.

We also explored whether the performance advantages for RRs were present only for those who had correctly identified articles as RRs or not. To the extent that certain characteristics of articles (for example, reporting style or content) are viewed as more rigorous and also are a good indicator of an article being a RR, it would not be surprising to see a relationship between the RR performance advantage and correctly guessing each type of article. A more

Table 1 | Descriptive statistics of the last study of RRs and comparison articles separated by author-based and journal-based comparisons

	RR	Author-based comparison	Journal-based comparison
N	29	28	29
Statcheck results (last study)			
Detectable statistical tests	12 (41%)	10 (36%)	12 (41%)
Errors			
Median	0	0	0
Mean (s.d.)	0.67 (1.07)	0.50 (0.71)	0.58 (1.00)
Paper with > 0 errors	4	4	4
Statistical significance decision errors			
Median	0	0	0
Mean (s.d.)	0.08 (0.29)	0.10 (0.32)	0.17 (0.58)
Papers with > 0 decision errors	1	1	1
Open practices (last study)			
Open materials	17 (59%)	3 (11%)	2 (7%)
Open data	25 (86%)	5 (18%)	4 (14%)
Preregistration	17 (59%)	2 (7%)	0 (0%)
Last study characteristics			
Sample size			
Mean	358.4	310.3	1,150.0
S.d.	427.4	639.9	5,119.0
Median	220	118	78
Sample justification (yes/no)	26/3	8/20	5/24
Difficult to reach population (yes/no)	1/28	6/22	6/23
Online data collection	13 (45%)	5 (18%)	4 (14%)
Experimental/non-experimental	24/5	18/10	22/7

Notes: Statcheck 'papers with detectable statistical tests' includes papers for which statcheck detected at least one statistical result, namely correlations and t, F, χ^2, Z and Q tests, reported in American Psychological Association style; 'errors' indicates all detected statistical reporting errors; 'decision errors' indicates results reported as statistically significant when recomputed results were not, or vice versa. In principle, all RRs were preregistered with the journal, but papers were not counted if a preregistration could not be found in an independent registry.

interesting question is whether even reviewers who did not correctly guess article type would still perceive improved quality and rigour. In the within-subjects model, 19 of 19 outcome criteria remained directionally consistent with an RR performance advantage for reviewers who guessed one paper correctly ($n = 168$), and 14 of 19 outcome criteria remained directionally consistent for reviewers who guessed neither paper correctly ($n = 87$). And, for both, the average raw effect sizes were weaker than observed in the aggregate data (one paper correctly guessed: mean 0.44, range −0.23 to 0.71; neither paper correctly guessed: mean 0.10, range −0.03 to 0.35; see Extended Data Fig. 4 and Supplementary Table 4 for more details). Perceived performance advantages of RRs are associated with noticing that the papers are RRs, but these results provide modest evidence that reviewers may perceive some performance advantages even if they do not notice or correctly guess that a paper is an RR.

Coding outcomes. We coded objective features of the articles that could be associated with greater rigour and transparency. Here, the unit of analysis is the article, substantially reducing precision because of the limited sample size. Table 1 summarizes descriptive evidence that statistical reporting errors as identified by statcheck[33] occurred infrequently. Preregistration occurred much more frequently with RRs, as should be expected because the format includes precommitment to design and planned analysis. Lack of preregistration means that a link to a publicly available precommitted design and analysis plan was not included in the paper for verifiability. Observing less than 100% preregistration indicates an opportunity for improving formal documentation of RRs[34,35]. Sharing materials and data were observed more frequently in RRs than comparison articles. Finally, median sample sizes were notably larger in RRs, and sample size justifications and online data collection were more common, but studies with difficult to reach populations were less common.

Discussion

Reviewers rated RRs similarly to or more positively than non-RR comparison articles across 19 outcome criteria (mean performance advantage 0.46, range 0.13–0.96). This advantage held whether considering a within-subjects design or a between-subjects design comparing whichever article reviewers examined first. It also held whether reviewers had experience or familiarity with RRs or preregistration, and was weaker but not reversed if they held equivocal beliefs about RRs in general. The performance advantage held among criteria that were assessed before showing the findings (mean 0.45, range 0.13–0.97), after showing the findings (mean 0.39, range 0.18–0.66) and after finishing the paper (mean 0.51, range 0.28–0.97). Finally, RRs investigated in this study were associated with larger median sample sizes, more sharing of data and materials, stronger presence of preregistration than comparison articles and somewhat less use of hard to reach samples.

RRs' performance advantage was strongest on the factors that should be most directly the basis of peer review: methodology and analysis rigour and quality. The RR performance advantage extended to characteristics of design such as alignment between the question and methodology and reporting such as alignment between the findings and interpretation. Sceptics have suggested that the planning required for RRs could undermine discovery, reduce novelty and creativity, promote boring research and undermine innovation[29,30]. We found no support for these speculations. For some criteria, the 95% credible intervals included 0, but for none of the 19 outcome criteria did the standard model numerically outperform RRs.

Limitations and constraints on generality. This comparative study provides important initial evidence about the quality and performance of RRs for research credibility. However, it is not a definitive comparative investigation of RRs versus the standard publishing model. The naturalistic design of this study means that there was no random assignment of research to be conducted as an RR or not. The naturalistic design also means that the included RRs were not performed by a random subset of authors or published in a random subset of journals, as authors and editors self-select whether to use RRs. The matching strategy we used was designed to reduce these concerns, minimizing confounding explanations by matching on topic as well as on journal and lead author. A randomized trial of RRs would be worth conducting, although it will be difficult to avoid researchers becoming aware of their participation and treatment group. Our naturalistic approach is arguably less subject to reactivity on the part of reviewers because the review is naturalistic and retrospective. Also limiting the generalizability of these results is the simple fact that the RR publication format is new. Behaviour during early adoption of publishing practices may not be the same as later adoption. Early adoption could suffer from initial weaknesses in RR implementation, or it could be particularly high quality due to extra attention based on its novelty. Simultaneously, the researchers and editors who are early adopters of the format may be different in important ways from later adopters. Another limitation is that we used peer review evaluation of research quality rather than other assessments of research quality, such as assessing whether RRs produce findings that are more replicable or generalizable. While such outcomes are difficult or resource-intensive to assess, they are necessary in combination with expert evaluation to get a complete understanding of research quality. Finally, we recruited reviewers and matched them to topics based on expertise similarly to standard peer review. However, we cannot estimate their representativeness of reviewers in general. Despite these limitations, this study provides a basis of evidence to expand the use of RRs in research, and should spur follow-up research to examine the generality of these findings. We do not have strong reasons to expect that these findings will fail to generalize to other fields, other methodologies and other reviewers. Nevertheless, these all deserve empirical scrutiny.

Conclusion

RRs were conceived to alter the incentives for authors and journals away from producing novel, positive, clean findings and towards conducting and publishing rigorous research on important questions[1–3]. Making acceptance decisions prior to knowing the outcomes should and does mitigate publication bias favouring positive results[26,27]. The present evidence indicates that RRs are associated with reviewers perceiving greater rigour and quality without costs on research importance or creativity compared with the standard model. We speculate that the higher quality of RRs is a function of three factors: authors put greater emphasis on defining their research questions and designing their methodologies to achieve acceptance, peer reviewers and editors are more likely to evaluate the methodology when there are no findings to distract or excuse poor methodological decisions and peer review can prompt study design improvements before conducting the research.

Shifting the incentives toward quality and rigour may have a variety of salutary effects on the published literature and research culture. The pressure to achieve positive results creates corrosive incentives that reward questionable research practices and undermine the credibility of findings[4,36,37]. Realigning reward systems with the values of quality and rigour could improve research credibility and accelerate the discovery of knowledge, solutions and cures. RRs are not a panacea, but the evidence suggests that they offer clear and tangible benefits to improving research and the research culture.

Methods

This research protocol was reviewed and approved by the University of Virginia Institutional Review Board for Social and Behavioral Sciences (protocol #3077). All participants consented to participate.

RR article selection. The study was conducted using RRs published from 1 January 2014 to 18 April 2018 that were identified from the 102 journals that offered RRs at that time either as a regular submission option or as part of a single special issue (https://cos.io/rr). An ongoing list of all published RRs is available in this database (https://www.zotero.org/groups/479248/osf/collections/KEJP68G9). For this study, we used RRs from psychology and neuroscience. We set these inclusion criteria for efficient implementation of the comparison matching process, for feasibility of defining a coding scheme that could be administered consistently across studies and for feasibility of recruiting peer reviewers. We also excluded RRs reporting replications. Replications are rare in the non-RR literature and popular with the RR format. Including replications in our RR sample would induce a confound as we would not be able to find enough matched non-RR replication studies. Supplementary Fig. 1 shows the effect of our inclusion and exclusion criteria on our article sample. In total, we included 29 RRs reporting novel research from psychology and neuroscience (Supplementary Table 5).

Comparison article selection. We used existing RRs that investigated novel research questions and a sample of comparison articles from the published literature. This has the advantage of ecological validity, but the disadvantage of being a quasi-experimental design that could introduce threats to construct validity of RRs versus the standard model. We preregistered two complementary, bias-minimizing matching procedures to select comparison articles for each RR. One procedure matched articles primarily based on journal and date of publication to emphasize comparability on publishing selectivity, selection criteria and journal style and standards. The other procedure matched articles primarily based on lead authors and date of publication to emphasize comparability of research style of those reporting the research. Each of the 29 RRs, with 1 exception, had matched articles for both selection criteria, resulting in 57 comparison articles (see Supplementary Information for details), for a total of 86 articles (Supplementary Table 5). This systematic matching process minimizes selection bias and maximizes the opportunity to observe variations in qualities of articles that are plausibly attributable to whether the article was an RR or not.

Article coding. We coded the last study of each article on objective criteria including sample size and sample justifications, whether participants were recruited from difficult to reach populations and study design (experimental or non-experimental). Coders also identified what they considered to be the study's primary outcome (see Supplementary Information for details). Two individual raters were randomly assigned to each article to independently apply the coding scheme for assessing objective characteristics of methodological rigour and reporting of outcomes. Agreement was computed using Krippendorff's alpha reliability coefficient with the 'irr' package in R. Agreement was 0.943 for sample size, 0.788 for sample justification, 0.782 for difficult to reach population and 0.576 for study design. In cases of disagreement, a third coder resolved any coding disagreements. We also coded whether articles reported studies that were preregistered, had open materials or open data, or collected data online. To do so, we scraped and analysed text from each article to identify terms referencing each variable, which were confirmed by an individual coder. We did not plan to conduct inferential statistics for the coding. Instead, we planned to report counts, means and standard deviations for coded variables across RR and comparison articles.

We also ran each paper through the R package statcheck[33,38] to identify errors in statistical tests reported in American Psychological Association style. Errors were those in which the reported statistical estimate, P value and degrees of freedom contained an inconsistency. Decision errors were those in which the statistical estimate was reported as statistically significant but the recomputed value was not, or vice versa.

Peer reviewer sample. We planned to obtain a sample size of 360 survey respondents (average of ~12 respondents per RR). This is equivalent to achieving 96% power to detect a medium effect size (Cohen's d of 0.4), assuming the between-article pair proportion of variance is similar to the between-lab proportion of variance observed in multi-lab studies (3.85%), or 80% power to detect the same effect size assuming a higher between-article proportion of variance (~10%; script: https://osf.io/fgkrs/). With our analysis strategy, we did not need to achieve perfect matching of sample size for each RR; however, we aimed for an even distribution of respondents across RRs while prioritizing maximizing expertise alignment. In Supplementary Table 6, we report the number of respondents per RR article pair.

We recruited reviewers with individual emails starting 16 October 2019 and ending 23 December 2019. The recruiting email invited them to participate in a study to peer review two published articles and evaluate qualities such as creativity and rigour on rating scales, a task that would take up to 2 h for compensation of $80. Potential reviewers were given a 3-week window from the date of invitation to complete the reviews. And, finally, to facilitate non-experts self-selecting out of participating, the recruiting email identified eligibility as researchers with expertise in the subdisciplines of the selected articles.

Our primary sample was a list of emails of 7,100 faculty in US-based psychology departments that had been constructed in 2017. With the passage of time, some of the email addresses were no longer valid due to faculty retirement or relocation. Also, a majority were in subdisciplines not represented in our sample of RRs, making them unlikely to meet eligibility criteria. Nevertheless, this sample of email addresses provided an unbiased approach to recruiting potential peer reviewers. To achieve our target sample size and maximize coverage of peer reviews across the selection of articles, we supplemented this primary recruiting strategy with additional directed recruiting of reviewers representing specific topical areas. We identified 31 journals in Web of Science that clearly published research related to the topics of our sample of papers. We then extracted 398 European email addresses and 337 US email addresses of corresponding authors from Web of Science from recent publications in those journals.

A total of 7,835 email invitations were sent, and 353 peer reviewers completed at least one outcome measure (4.5%). We do not know the breakdown of reasons for declining to participate, but almost all of the more than 100 spontaneous 'decline to participate' responses indicated that the specific disciplinary topics were outside their expertise, they did not have time to do the reviews in the next 3 weeks or they had retired and were no longer reviewing. Participant reported gender was 56% male, 42% female and 2% preferred not to say. We did not assess participant age.

Prior to consenting for the study, potential reviewers completed a short eligibility assessment to determine whether their areas of expertise matched any of the articles in the sample. Potential reviewers first identified the subdiscipline closest to their area of expertise (social-personality psychology, cognitive psychology, neuroscience or none of the above). If they selected 'none of the above' they were informed that they did not match any articles and thanked for their interest in participating. If they selected social-personality psychology they were presented with five options (media/applied psychology, group processes and intergroup relations, positive psychology, personality psychology or attitudes and social cognition). If they selected a disciplinary area matching our set of articles, they were presented with article keywords in sets and identified the set of keywords that most closely matched their expertise. The article keyword sets were constructed by combining the keywords from an RR and its two matched comparison articles. This approach enabled us to match researchers based on expertise without revealing the selected articles and without biasing selection by expertise on the RR or comparison articles.

After selecting a keyword set, potential reviewers consented to participate and then were assigned to review the selected RR and a randomly selected comparison article from the pair. The two articles were presented in a randomized order via Qualtrics, and reviewers completed one review before proceeding to the next article. Reviewers could pause and return to complete the reviews if they could not complete the task in a single sitting. A total of 353 reviewers were matched to articles and completed at least one outcome measure. Because we matched on expertise, the number of reviewers was not equally distributed across articles. Of the 29 RR and control article matches, each had an average of 12.5 reviewers with a low of 3 and a high of 24 (s.d. 5.59).

Structured peer review survey. We developed a structured peer review survey to evaluate the papers on 19 qualities. Each reviewer completed a subjective evaluation of two papers—an RR and a control paper randomly selected from the RR's two comparison papers. Papers were lightly redacted to remove any incidence or indication of the words 'registered report'. We expected that at least some papers and some reviewers would still be able to identify RRs. We considered trying to completely blind the studies. However, in test cases we found that this resulted in removal of content that was directly relevant for evaluating rigour and transparency of the study design and findings. At the end of the study, we asked reviewers whether each of the papers was published as an RR. We also assessed reviewers' familiarity with and beliefs about RRs in general.

To minimize the influence of the findings on evaluation of the methods, the content of each paper was presented and reviewed in the following fixed order: (1) introduction, initial studies, and methods of the last study; (2) results of the last study and discussion; (3) abstract. Section headings, text, tables, figures and references were included, while other paper attributes (for, example author(s) and journal) were not included. Reviewers completed all measures for one section before proceeding to the next section.

Reviewers provided quantitative responses on bipolar nine-point Likert scales ranging from −4 (substantially less than the average study) to 4 (substantially more than the average study), with 0 indicating equivalence to 'the average study [article]'. The average study reference point is important for encouraging full use of the scale rather than making an absolute assessment of 'meets publishing criterion' as is typical for peer review decisions to accept or reject papers for publication. Each section included a short open-response box if reviewers wished to explain their responses. These responses were retained for exploratory purposes. All survey questions focused on the last study of the paper unless specifically stated to comment on the paper as a whole.

Preregistered analyses. We were interested in the causal effect of an article being an RR on our survey questions. Our data are nested: Reviewers are nested within article pairs, and article pairs are nested within discipline. For each article pair, there is an effect of article type on each survey response. The overall effect of interest is the average effect of article type across article pairs. This is similar to a meta-analytic effect, with the article pairs being similar to effects from individual studies. We used multi-level modelling to account for our nesting and to allow the effect of article type to vary across the different article pairs, partially pooling the estimates to create an overall average effect. In our preregistration, responses to the survey questions were modelled separately, running a mixed-effects model for each outcome, as we did not have strong a priori predictions about higher-order latent constructs and so were not sure whether the assumption that would be required to run all outcomes in a single model would hold. Data analysis was not performed blind to the conditions.

Within versus between effects. For each article pair, there were two ways we can calculate the effect of article type: (1) a within-subjects comparison using reviewers' responses to both articles or (2) a between-subjects comparison using reviewers' responses to the randomly assigned first article. Between-subjects effects have less power than within-subjects effects, but we preregistered both within- and between-subjects analysis to address two concerns. First, a priori we were concerned that the length of our survey may lead to high drop-out after the first article, potentially hampering the within-subjects model. In reality, 99.43% of the reviewers provided data for both articles. Second, even if we did not have high attrition, a between-subjects comparison minimizes reviewers' awareness of the treatment. After the first article they have only experienced one condition. For these reasons, we estimated both a within-subjects and between-subjects effect of article type by running two different models for each survey question.

Crossed effects. Once reviewers were placed into an RR condition using keywords, they were then randomly assigned to matching condition (which of two comparison articles, same author or same journal) and a counterbalance condition (which of two orders to review the articles, RR first or second). This means that each article pair is crossed with matching and counterbalanced conditions. This allows us to estimate variation across matching and counterbalance conditions.

Model 1: within-subjects model. In this model, we estimated the overall within-subject effect of article type on each of our survey items. In our original preregistration (https://osf.io/a6xfj), we stated that we would include fixed effects for article type, matching scheme, counterbalance condition, the four interaction terms between these three predictors and the discipline of each article pair. We also stated that we would include random effects for subject to account for the within-subjects design, an intercept for article pair and a random slope for article type by article pairs, and that our outcome measure would be a difference score. This model is impossible to estimate (as the difference score removes article type as a possible effect to estimate), and we realized this mistake when analysing the first outcome. At that point, we updated the preregistration (https://osf.io/wjxsg) based on a comparison of fit between two different potential specifications of the within-subjects model, and the model that fit better was then used for all other outcomes (which had not been looked at when the preregistration was amended). This new model included a difference score outcome, fixed effects for article discipline, matching scheme, counterbalance condition and the interaction between matching and counterbalance, and a random intercept for article pairs. The intercept of this model estimates the article type effect. See Supplementary Table 7 for full model specification.

Model 2: between-subjects model. In this model, we estimated the overall between-subject effect of article type on each of our survey items, using only survey responses to the first article rated by each reviewer. We modelled each survey response item as a function of the discipline, article type, matching scheme and interaction of article type and matching scheme. These four terms plus the intercept resulted in a total of five fixed-effect terms in the model. We included a random intercept for the article pair and a random slope of the article type by article pair. See Supplementary Table 8 for full model specification.

Bayesian estimation methods. We use Bayesian methods to estimate the mixed-effects models. We assigned uninformative priors to all fixed-effect parameters so that the results depend only on the data, similar to frequentist inference methods. One analyst (C.K.S.) ran the models in Stan (using the brms R package) with Hamiltonian Monte Carlo methods with four chains[39]. A second analyst (K.M.E.) independently reproduced the results of Fig. 1 in OpenBUGS, and the reproduced results are identical to those reported herein.

We set weakly informative priors for both the between-subject and the within-subject model. We set priors of $\mathcal{N}(0, 2)$ for the coefficients because the response scale is a nine-point scale ranging from −4 to 4 for fixed effects. Setting the standard deviation of the prior to 2 is conservative in that it places much of the likelihood around smaller changes in rating across conditions. And, we used half-$\mathcal{N}(0, 2.5)$ priors for the priors of the hyperparameters to achieve partial pooling. We also performed sensitivity analyses to assess the extent to which our priors are affecting our estimates and found that the priors we use have no relevance to the results. For example, in the reproduction, we reparameterized the standard deviation parameters using the exponential function, and then assigned flat $\mathcal{N}(0, 1{,}000)$ priors for all parameters but found identical results.

We used a number of different procedures to assess model fit and to check for convergence. To assess convergence we visually inspected the trace plots and Rhat values. Following evidence of model convergence, we assessed the fit of the models to the data through posterior predictive distributions. We created posterior predictive distributions for the global (overall average) article type parameter, its standard deviation and the maximum and minimum values for article pairs. We compared these simulated values with the observed values in our dataset to determine whether the model is accurately capturing different important facets of our data. For all models, we report the widely applicable information criterion (WAIC) (or leave-one-out (LOO) if suggested in place of WAIC by the Bayesian estimation program), parameter estimates and 95% credible intervals.

Preregistration availability. The preregistered design and analysis plan is available on OSF: https://osf.io/a6xfj and https://osf.io/wjxsg

Reporting summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability
All data files are available on OSF: https://osf.io/aj4zr/.

Code availability
All files and scripts are available on OSF: https://osf.io/aj4zr/.

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Author contributions
Conceptualization: Survey: C.K.S., T.M.E. and B.A.N. Article coding: S.R.S., J.B. and S.V. Data curation: Survey: C.K.S. Article coding: S.R.S. and J.B. Formal analysis: Survey: C.K.S. and K.M.E. Article coding: S.V., S.R.S. and J.B. Investigation: Survey: C.K.S. and T.M.E. Article coding: S.R.S., J.B. and S.V. Methodology: Survey: C.K.S., T.M.E., K.M.E. and B.A.N. Article coding: S.R.S., J.B. and S.V. Software: Article coding: S.R.S. and J.B. Visualization: Survey: C.K.S. Article coding: S.R.S. and J.B. Validation: Survey: K.M.E. Article coding: S.V. Project administration: T.M.E. Resources: T.M.E. and F.S.T. Supervision: T.M.E. and B.A.N. Funding acquisition: T.M.E. and B.A.N. Writing, original draft: C.K.S., T.M.E., K.M.E. and B.A.N. Writing, review and editing: C.K.S., T.M.E., S.R.S., J.B., F.S.T., S.V., K.M.E. and B.A.N.

Competing interests

T.M.E., C.K.S. and B.A.N. are employees of the nonprofit Center for Open Science (COS), which has a mission to increase openness, integrity and reproducibility of research. COS offers support to journals, editors and researchers in adopting and conducting RRs. The remaining authors declare no conflicts of interest.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s41562-021-01142-4.

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41562-021-01142-4.

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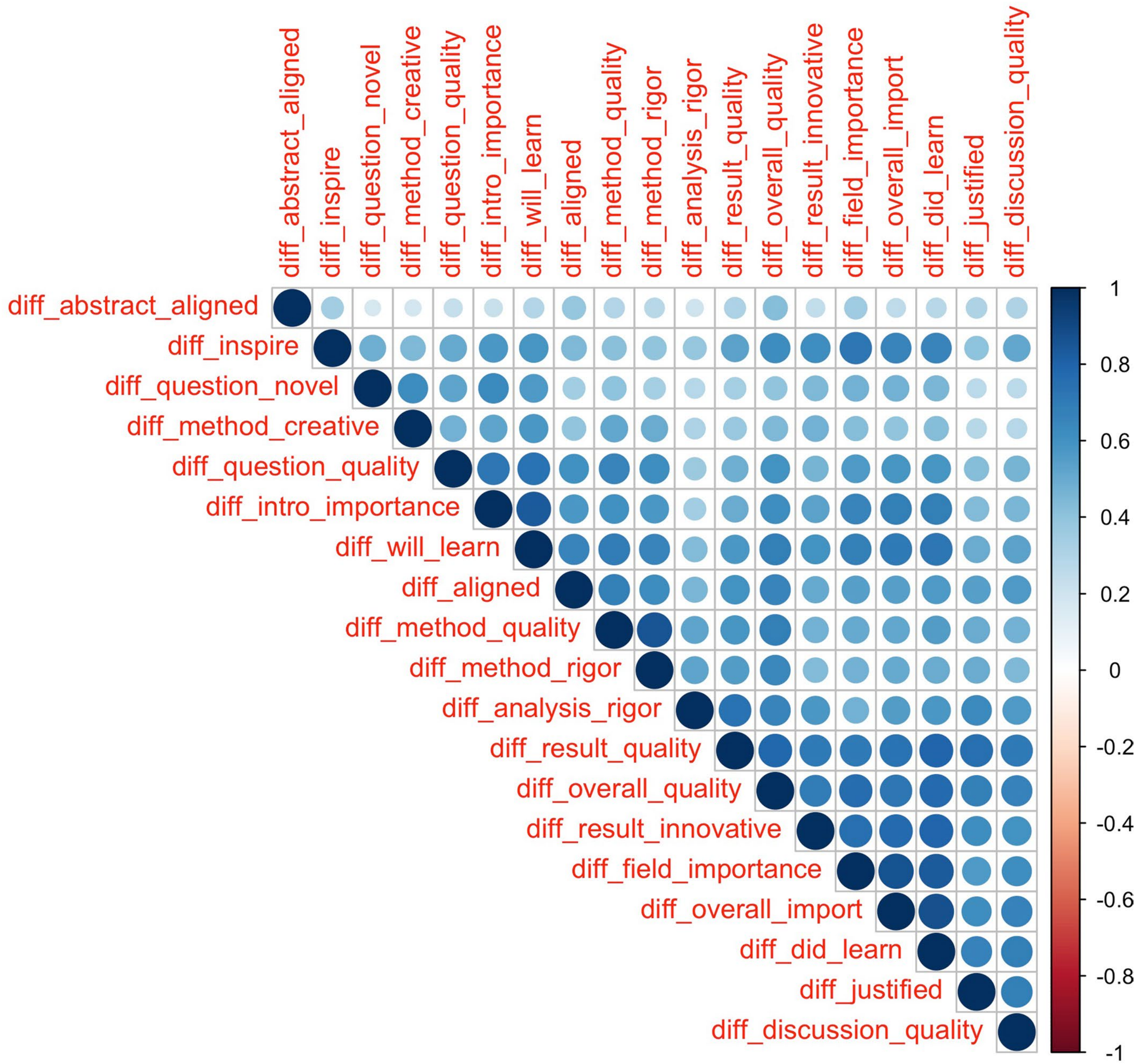
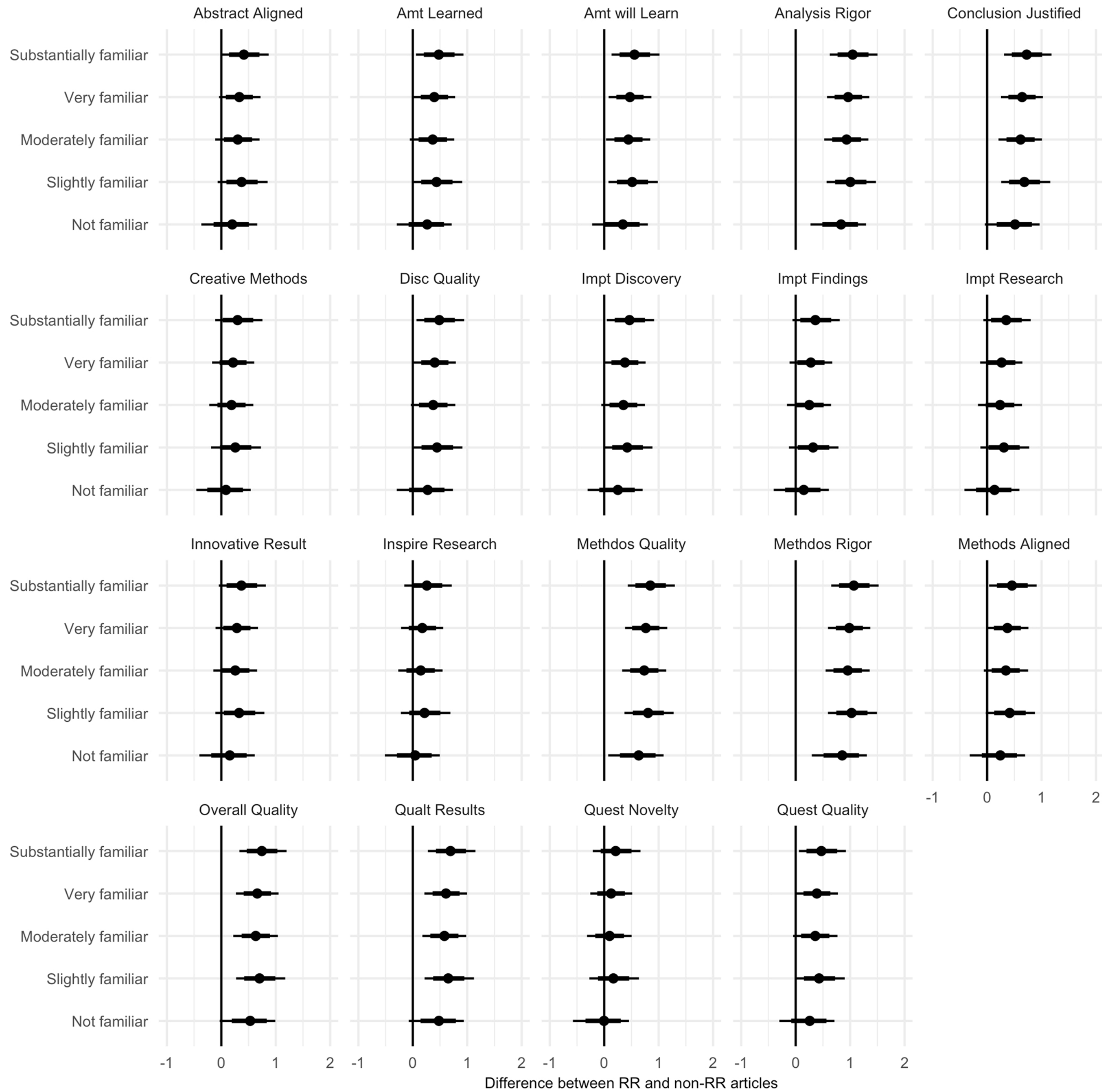
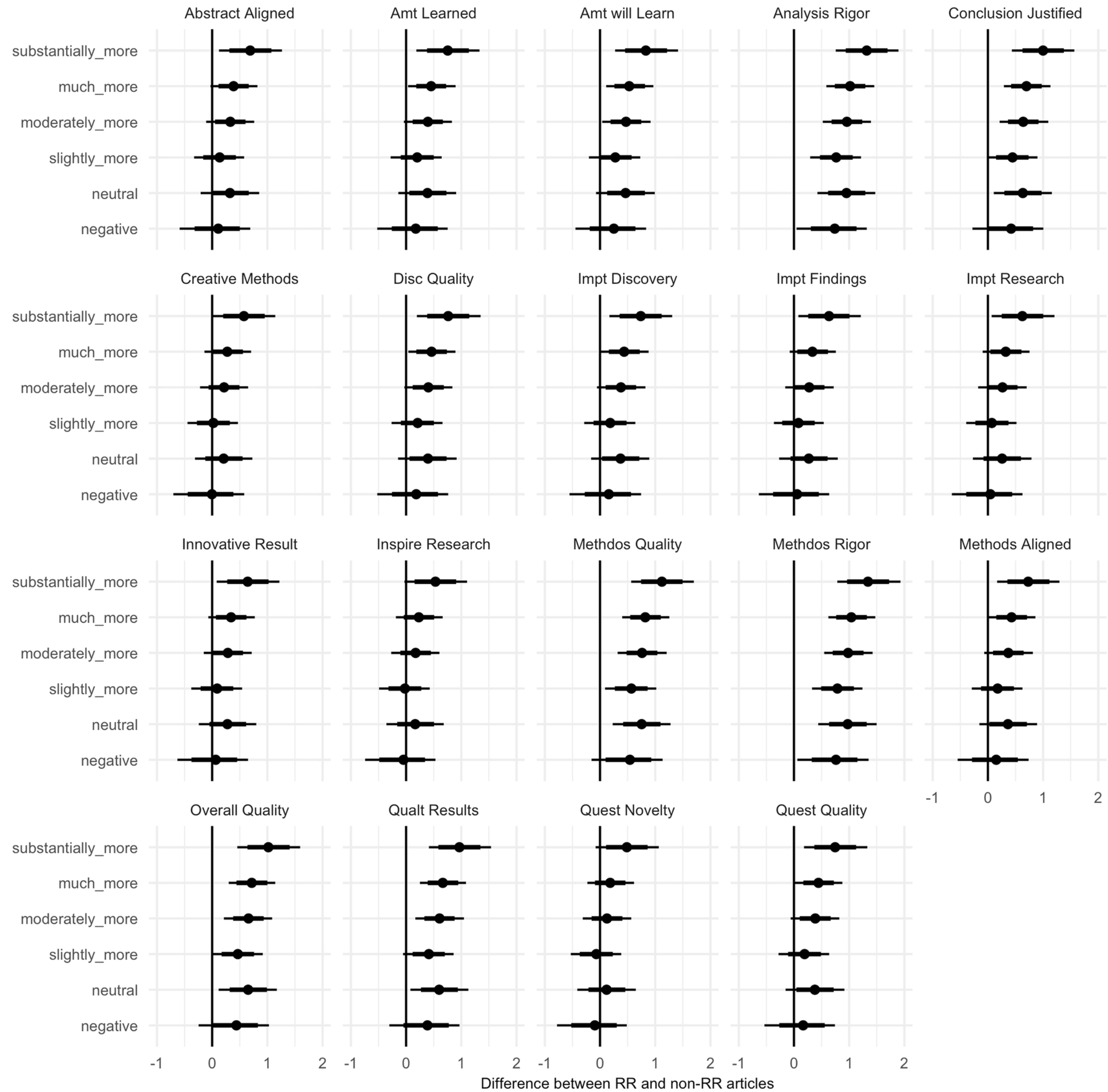
Peer review information *Nature Human Behaviour* thanks Balazs Aczel, Marcel van Assen and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.



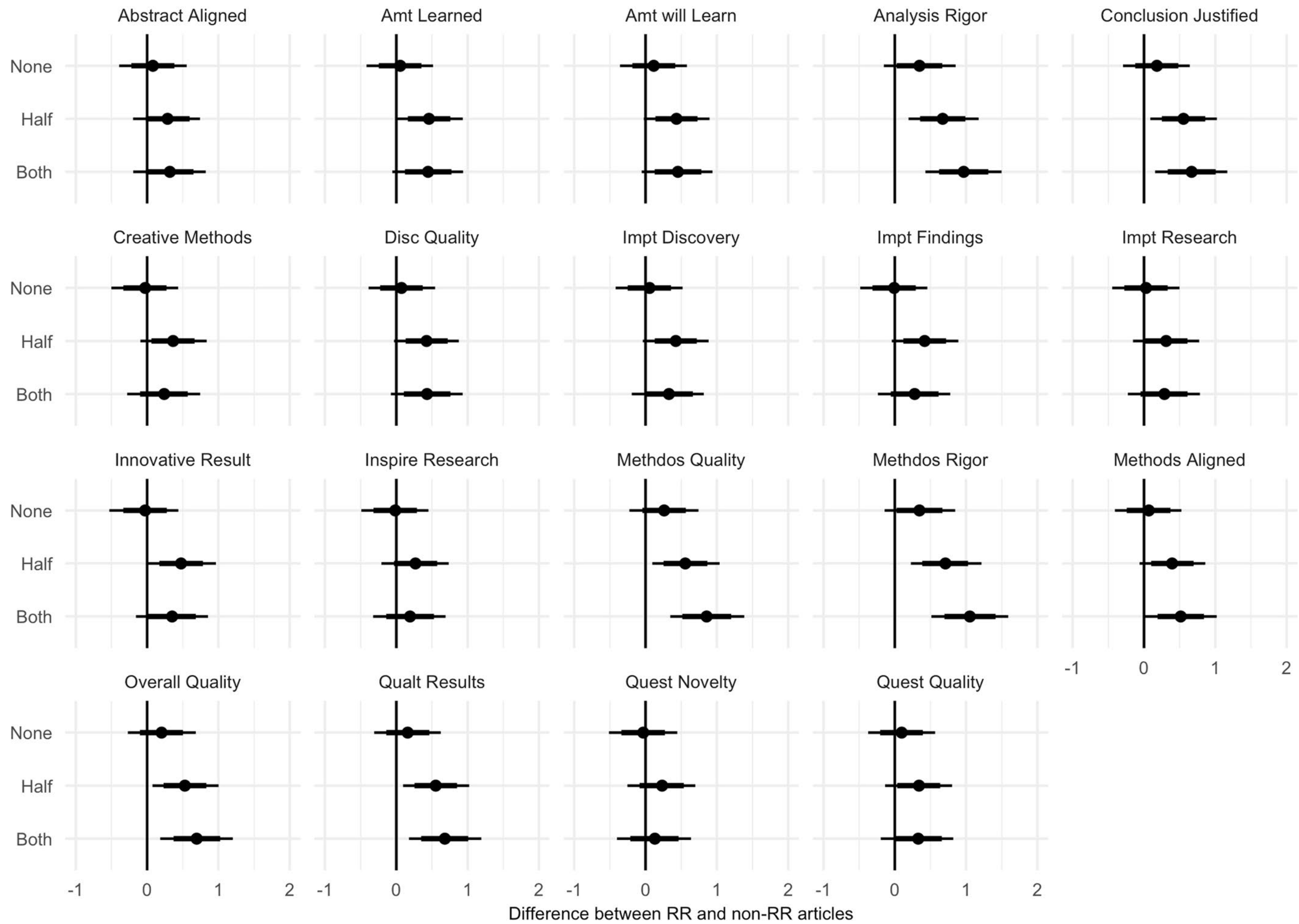
Extended Data Fig. 1 | Plot of correlations between all difference score outcome variables. Correlation matrix of the 19 outcome variables with larger darker blue circles indicating stronger positive correlations than smaller lighter blue circles.



Extended Data Fig. 2 | Posterior probability distributions for parameter estimates for each DV and each level of Familiar comparing the difference of RRs and comparison articles. Horizontal lines indicate 80% (thick) and 95% (thin) credible intervals and dots show the mean of the posteriors. Positive values indicate a performance advantage for RRs, negative values indicate a performance advantage for comparison articles.



Extended Data Fig. 3 | Posterior probability distributions for parameter estimates for each DV and each level of Improve comparing the difference of RRs and comparison articles. Horizontal lines indicate 80% (thick) and 95% (thin) credible intervals and dots show the mean of the posteriors. Positive values indicate a performance advantage for RRs, negative values indicate a performance advantage for comparison articles.



Extended Data Fig. 4 | Posterior probability distributions for parameter estimates for each DV and each level of 'Guessed Right' comparing the difference of RRs and comparison articles. Horizontal lines indicate 80% (thick) and 95% (thin) credible intervals and dots show the mean of the posteriors. Positive values indicate a performance advantage for RRs, negative values indicate a performance advantage for comparison articles.

nature research

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Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Data was collected using Qualtrics.
Data analysis	Data was analyzed using R version 3.6.0

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Behavioural & social sciences study design

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Study description

Naturalistic quasi-experimental design, quantiative methods

Research sample

A total of 7,835 email invitations were sent and 353 peer reviewers completed at least one outcome measure (4.5%) for the 86 papers in our sample (29 published RRs from psychology and neuroscience and 57 non-RR comparison papers). Recruited reviewers were matched to topics based on expertise similarly to standard peer review. However, we cannot estimate their representativeness of reviewers in general.

Sampling strategy

Convenience sampling.

The study was conducted using Registered Reports (RRs) published from January 1, 2014 to April 18, 2018 that were identified from the 102 journals that offered RRs at that time either as a regular submission option or as part of a single special issue (https://cos.io/rr). We preregistered two complementary, bias-minimizing matching procedures to select comparison articles for each RR. One procedure matched articles primarily based on journal and date of publication to emphasize comparability on publishing selectivity, selection criteria, and journal style and standards. The other procedure matched articles primarily based on lead authors and date of publication to emphasize comparability of research style of those reporting the research.

Our primary sample for peer reviewers was a list of emails of 7,100 faculty in U.S.-based psychology departments that had been constructed in 2017. We supplemented this with corresponding authors from Web of Science from recent publications in journals publishing related topics as our sample of papers. 398 European email addresses and 337 U.S. email addresses were extracted.

We planned to obtain a sample size of 360 survey respondents (average of ~12 respondents per RR). This is equivalent to achieving 96% power to detect a medium effect size (Cohen's d of 0.4), assuming the between-article pair proportion of variance is similar to the between-lab proportion of variance observed in multi-lab studies (3.85%), or 80% power to detect the same effect size assuming a higher between-article proportion of variance (~10%; script: https://osf.io/fgkrs/).

Data collection

Online structured peer review survey to evaluate the papers on 19 qualities. Each reviewer completed a subjective evaluation of two papers -- an RR and a control paper randomly selected from the RR's two comparison papers. Papers were lightly redacted to remove any incidence or indication of the words "Registered Report."

Timing

Registered Reports (RRs) published from January 1, 2014 to April 18, 2018 that were identified from the 102 journals that offered RRs at that time either as a regular submission option or as part of a single special issue (https://cos.io/rr).

We recruited reviewers with individual emails starting October 16, 2019 and ending December 23, 2019.

Data exclusions

Because replications are rare in the standard model, conflating RRs with replications would interfere with the project aims. As such, we excluded replication RRs from this study and only examined RRs that investigated novel research questions. Figure S1 shows the effect of our inclusion and exclusion criteria on our article sample of 29 RRs reporting novel research from psychology and neuroscience (Table S1).

In total 13 participants who answered at least 1 question were excluded because they were an author on at least one of the papers they read about, in line with our preregistration (https://osf.io/a6xfj). Two were excluded because they explicitly stated that they had not followed directions (one specified that they had based their ratings on the first, rather than the last study; one specified that they couldn't scroll through the pdf and so had marked random responses). Eight were excluded because they were assigned to read a RR that was later determined to not actually be a registered report.

Non-participation

Our response rate was 4.5%

Randomization

Reviewers were assigned to the RR-comparison articles combo that they selected and then randomly assigned to which comparison article they would review. Reviewers completed the survey measures for one article before being presented with the survey measures for the second article. Reviewers were randomly assigned to receive the RR or matched comparison article first.

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☒	☐ Dual use research of concern

Methods

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☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Human research participants

Policy information about studies involving human research participants

Population characteristics

See above.

Recruitment

A total of 7,835 email invitations were sent with a link to the Qualtrics survey. We aimed to minimize selection bias of authors selecting articles that they know or wish to read by having reviewers identify their expertise through keyword matches. It is still possible however that reviewers were already familiar with the article. Similarly, we expected that at least some reviewers would still be able to identify RRs. At the end of the study, we asked reviewers whether each of the papers was published as an RR. We also assessed reviewers' familiarity with and beliefs about RRs or preregistration in general.

Ethics oversight

The IRB protocol was approved by the University of Virginia (protocol # 3077)

Note that full information on the approval of the study protocol must also be provided in the manuscript.